

Neural networks for groundwater prediction: a global literature review

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ABSTRACT

Groundwater is essential for water and food security, yet its prediction remains challenging because aquifer systems are nonlinear and monitoring data are often incomplete. This study aimed to determine the global state of the art in neural network applications for groundwater prediction through a systematic literature review. Publications from 2009 to May 2025 were retrieved from Scopus and Web of Science, yielding 359 records, of which 150 met the inclusion criteria. Data were organized and analyzed using Microsoft Excel, while VOSviewer 1.6.19 was used to characterize publication trends and keyword co-occurrence. India and Iran showed the highest scientific output. Long short-term memory (LSTM) models and hybrid architectures generally reported the strongest predictive performance, with several studies achieving root mean square error (RMSE) below 0.1 m and coefficients of determination (R^2) above 0.95. Wavelet, empirical mode decomposition (EMD), and variational mode decomposition (VMD), together with satellite observations, improved performance in data-limited settings. The review indicates that future research should prioritize interpretable, transferable, and computationally efficient models that integrate neural networks with hydrogeological principles to support sustainable groundwater management.

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1. INTRODUCTION

Groundwater represents one of the planet's primary sources of freshwater and is essential for human consumption, agriculture, and natural ecosystems [1]. More than 2.5 billion people depend directly or indirectly on groundwater to meet their basic needs [2]. However, climate change, population growth, and rapid urban expansion are placing increasing pressure on groundwater resources [3], [4]. Globally, aquifers supply approximately 38% of agricultural irrigation and nearly 50% of drinking water, making their effective and sustainable management an urgent necessity [5]. This challenge is particularly relevant in developing regions, where rural and peri-urban communities often depend heavily on wells and springs for water access

[6]. Intensive aquifer exploitation in regions such as South Asia, North Africa, and Latin America has contributed to declining water tables, salinization, and reduced water availability [7].

Groundwater monitoring and prediction have traditionally relied on physically based mathematical models governed by groundwater flow equations [8]. Although these approaches provide an important representation of aquifer processes, they require detailed hydrogeological information and may have difficulties representing the nonlinear behavior and temporal variability of groundwater systems [9]. These limitations have encouraged the application of artificial intelligence (AI) and data-driven approaches. In particular, artificial neural networks (ANNs), convolutional neural networks (CNNs), recurrent neural networks (RNNs), long short-term memory (LSTM) networks, and related machine learning approaches have demonstrated potential for identifying complex relationships within historical, meteorological, and hydrological data [10]–[12].

The existing literature can be broadly organized into three methodological groups. The first group comprises traditional ANN approaches, which have been applied to groundwater-level prediction and related hydrogeological variables, with several studies reporting prediction accuracies exceeding 95% for monthly groundwater-level estimates [13]–[16]. The second group includes recurrent and deep-learning architectures, particularly LSTM-based models, which are designed to capture temporal dependencies in groundwater time series and have demonstrated strong generalization performance [17]. The third group comprises hybrid and combined approaches that integrate neural networks with complementary machine learning algorithms or preprocessing techniques. Examples include adaptive neuro-fuzzy inference system (ANFIS), random forest (RF), and extreme gradient boosting (XGBoost)-based approaches, which have been used to improve predictive performance under different environmental and data conditions [18]–[20]. In addition, recent studies have increasingly incorporated preprocessing techniques and alternative data sources to address noise, nonstationarity, and data limitations.

Despite these advances, the existing literature presents several limitations. Previous reviews have examined the application of AI in hydrology [21], [22], but their scope has generally been broad and has not focused specifically on groundwater prediction. A more targeted review summarized machine learning methods applied to groundwater prediction [23]; however, it was limited to the 2010–2020 period and did not cover emerging architectures such as LSTM-CNN hybrids or transformer-based approaches. Furthermore, most individual investigations remain local in geographic scope and employ heterogeneous methodologies, input variables, preprocessing procedures, and evaluation metrics. This methodological heterogeneity makes comparisons across studies difficult and limits the identification of approaches that are robust and transferable across different hydrogeological settings.

The main research gap addressed by this review is therefore the lack of a global and structured synthesis that jointly considers neural network architectures, preprocessing strategies, evaluation metrics, and hydrogeological contexts in groundwater prediction. Addressing this gap is important because the rapid expansion of studies in this field has produced a fragmented body of evidence, making it difficult to determine which methodological approaches consistently provide reliable performance and under what data and hydrogeological conditions. A comprehensive synthesis is consequently needed to identify research trends, methodological strengths and limitations, inconsistencies among studies, and areas requiring further investigation.

The main contribution of this study is a global, structured synthesis of the literature on neural network-based groundwater prediction. The review covers publications from 2009 to 2025 and examines the evolution of research, geographical distribution, neural network architectures, preprocessing techniques, performance metrics, and the integration of hydrogeological and remote-sensing information. In addition, the review identifies emerging methodological trends and persistent research gaps, providing a basis for comparing approaches and identifying promising directions for future research. These findings are relevant both scientifically, by improving the understanding of methodological patterns and limitations in the field, and practically, by supporting the development of more interpretable, transferable, and computationally efficient prediction approaches for sustainable groundwater management.

Accordingly, the objective of this study was to determine the global state of the art in the application of neural networks for groundwater prediction, with emphasis on recent advances, methodological approaches, performance evaluation, key challenges, and their potential contribution to the sustainable management of groundwater resources.

2. METHOD

The study was conducted as a systematic literature review covering publications from 2009 through May 2025. The starting year is justified by the significant increase, beginning in 2009, in the systematic application of ANNs and other machine learning techniques in hydrology, coinciding with the expansion of deep learning in water sciences. The review process consisted of identifying relevant publications, screening

the retrieved records, applying the inclusion criteria, and synthesizing the eligible studies. The search was conducted using the terms “neural networks,” “prediction,” “groundwater,” and “sustainable management.” The investigation was conducted using the Scopus and Web of Science databases, which were selected based on their academic reputation and broad coverage of peer-reviewed scientific literature [24]. The search was limited to documents classified as article and review, with no additional subject-area filters applied. As a result, 143 articles were identified in Scopus and 216 in Web of Science; of which 67 and 83 studies, respectively, met the inclusion criteria and research objectives and were incorporated into the present review (Figure 1).

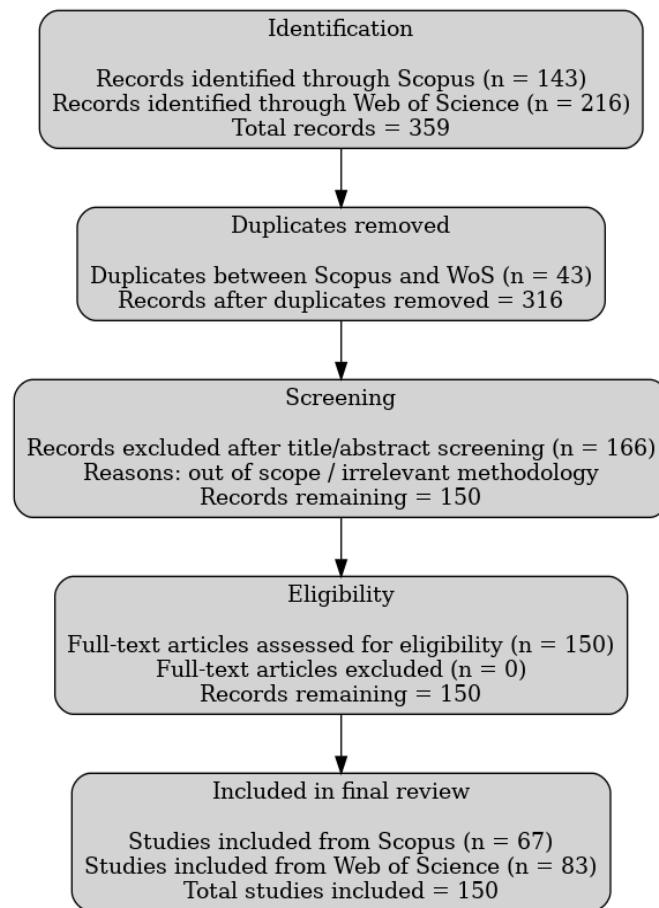


Figure 1. Flow diagram of the systematic review process

2.1. Inclusion and exclusion criteria

Publications from 2009 through May 2025, in all languages, were considered. Titles, abstracts, methodologies, and main findings were carefully examined to select articles of interest. The geographic scope was global. Documents such as book chapters, conference papers, and letters to the editor were excluded, as was grey literature because it is not subject to peer review. In addition, inconclusive studies and duplicate records were not taken into account (Figure 2).

2.2. Data analysis

The data were downloaded in CSV format and organized and systematically coded using Microsoft Office Excel. The extracted information was used to determine the distribution of studies by year and country, as well as the thematic distribution of prediction objectives. VOSviewer version 1.6.19 was used for keyword co-occurrence analysis. The analysis used all keywords as the unit of analysis and full counting as the counting method. A minimum occurrence threshold of five was applied, and no thesaurus file was used.

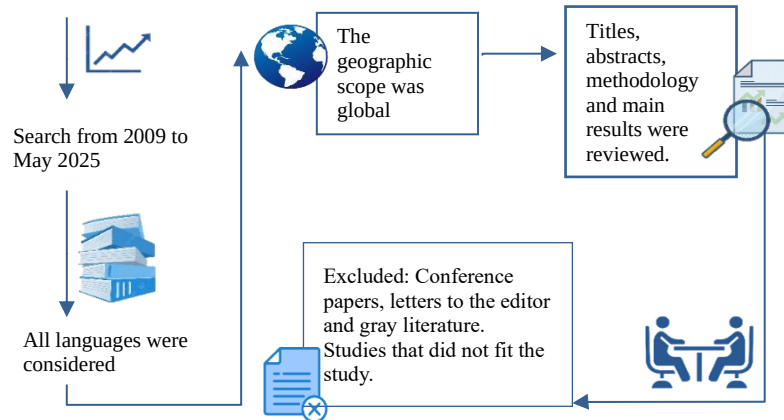


Figure 2. Exclusion and inclusion process

3. RESULTS AND DISCUSSION

Figure 3 shows the distribution of articles by author affiliation and country of origin, revealing that the nations leading research on the application of neural networks for groundwater prediction are India (29 articles) and Iran (24 articles). These findings are closely tied to the heavy reliance on groundwater resources in India, where much of the rural population depends on these sources as the primary supply for both domestic consumption and agricultural irrigation [25]. It is estimated that over 60% of all water withdrawals in India come from aquifers, making it the world's largest user of groundwater exceeding the combined withdrawals of China and the United States [26], [27]. This situation has raised concerns over falling water tables and has driven research initiatives aimed at monitoring and predicting these changes with greater accuracy [28]. Beyond this structural dependence, sociotechnical factors also play a role: in both India and Iran, the magnitude of the water crisis has motivated government-prioritized research programs and targeted funding in applied hydrology and machine learning, fostering a high level of scientific output [29], [30].

Iran, for its part, has faced a long-standing water crisis, relying increasingly on deep aquifers, particularly in agricultural and urban areas where precipitation is scarce and natural recharge is insufficient to replenish extracted water [31]. Groundwater has thus become an indispensable resource for sustaining both human consumption and agricultural activities, especially in the country's arid regions, where overexploitation has generated growing concern [32], [33].

The annual trend in publications also reveals steady growth beginning in 2009, reaching a peak of 44 articles in 2024 (Figure 4). This upward trajectory reflects the growing interest in applying neural networks to groundwater modeling and prediction, a resource that is increasingly critical in the context of climate change and water security [34].

Most of the reviewed scientific output focuses on studies whose primary prediction target is groundwater level (GWL), followed by research centered on groundwater quality (GWQ), groundwater potential (GWP), and, to a lesser extent, studies addressing the combined prediction of level and quality (GWL+GWQ). These topics underscore the current priorities in the application of neural networks to groundwater research (Figure 5).

Figure 6 presents the main predictive modeling techniques employed in the reviewed studies. The results show that ANNs of multilayer perceptron (MLP) and feed-forward neural networks (FFNNs) stand out as the most widely used architecture within the ANN family, followed by LSTM networks and support vector machines/regression (SVM). This pattern reflects a clear preference for supervised learning approaches focused on regression. Models such as RF and CNNs also exhibit a significant presence.

Figure 7 illustrates the frequency of different metrics used to evaluate neural network models predicting GWLs. The most commonly applied metric is root mean square error (RMSE), cited in over 100 of the reviewed studies. This prevalence indicates that many researchers favor RMSE for assessing model accuracy. Other widely used metrics include the coefficients of determination (R^2) and mean absolute error (MAE), which appear in approximately 70 and 60 studies, respectively. The Nash–Sutcliffe efficiency (NSE) coefficient and the correlation coefficient (R) also show significant use, while metrics such as mean squared error (MSE), mean absolute percentage error (MAPE), and relative absolute error (RAE) occur less frequently, with fewer than ten mentions each.

The dominance of RMSE highlights its ability to heavily penalize large errors, a critical factor in hydrology where extreme deviations in GWLs can have significant impacts (due to its quadratic component, RMSE amplifies large discrepancies) [35]. In contrast, MAE provides a more straightforward interpretation

of average error easy to express in physical units while R^2 measures explained variance, though it can be misleading if systematic biases or variability not captured by variance alone are present [36]. Other metrics offer complementary perspectives: for instance, NSE is widely valued in hydrological studies because it allows comparison of model performance against the observed mean.

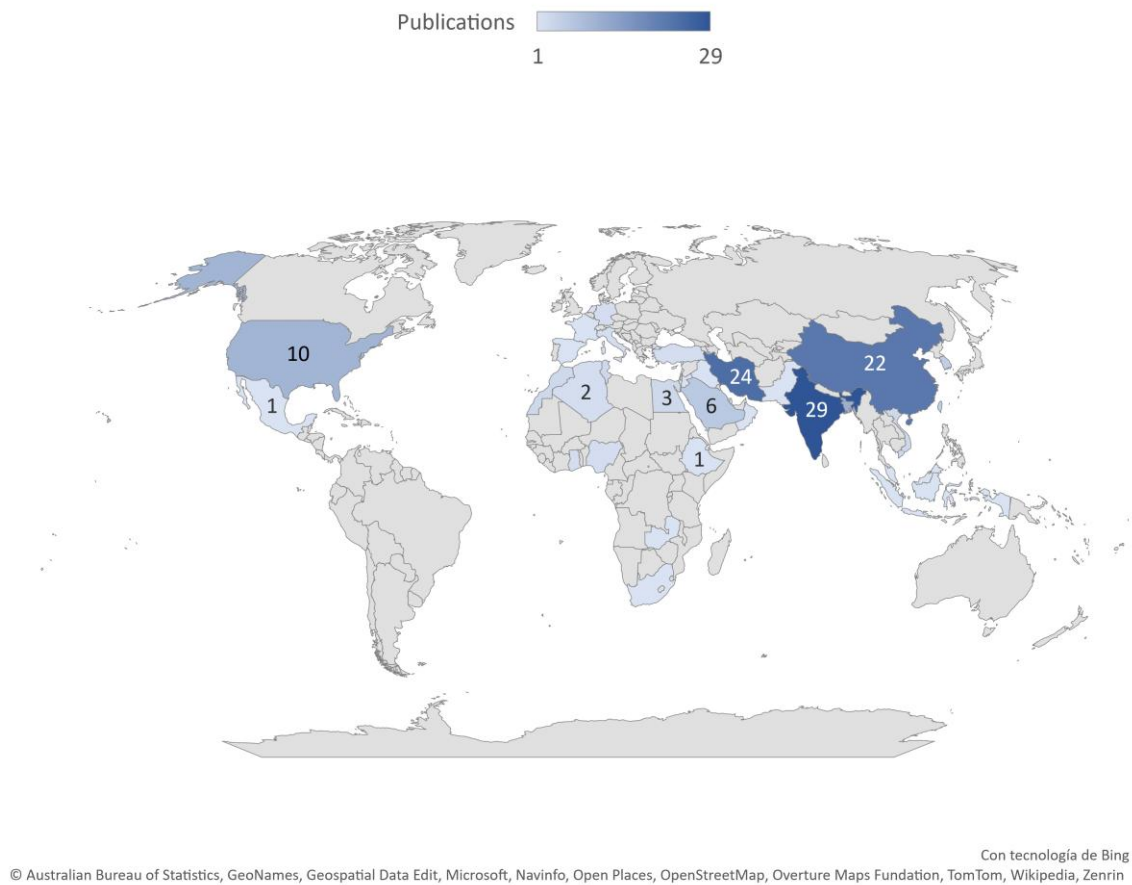


Figure 3. Distribution of scientific production by country

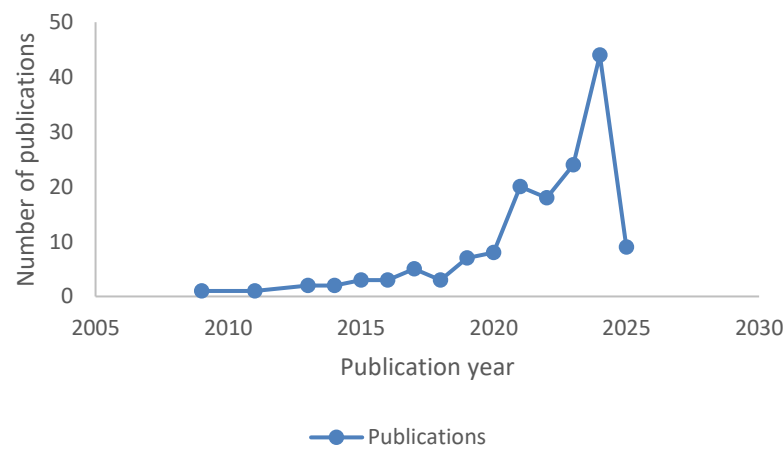


Figure 4. Annual distribution of publications in groundwater prediction (2009–2025)

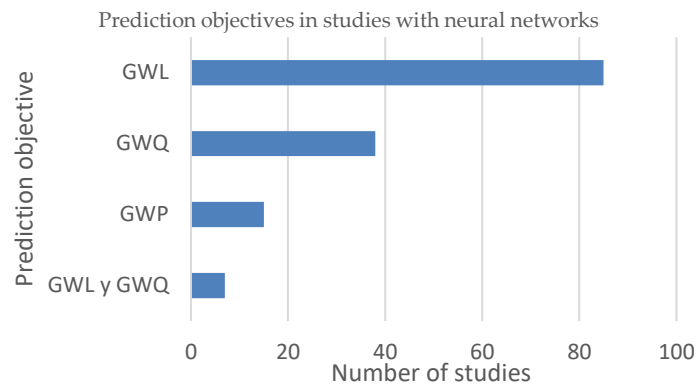


Figure 5. Thematic distribution of GWL, GWQ, and GWP

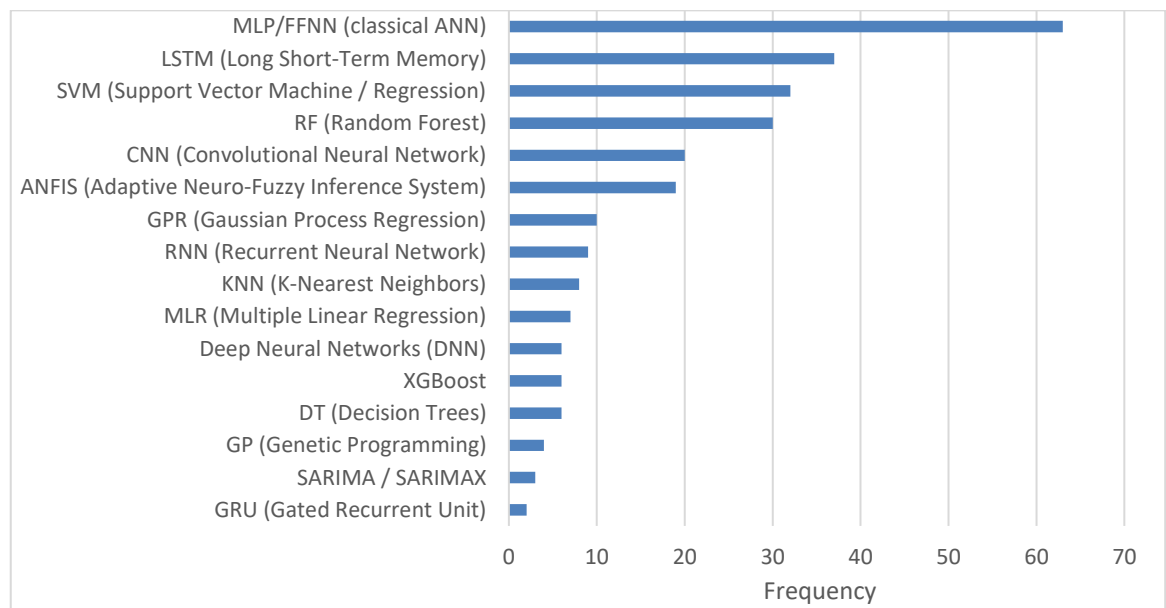


Figure 6. Known neural network architectures used in research

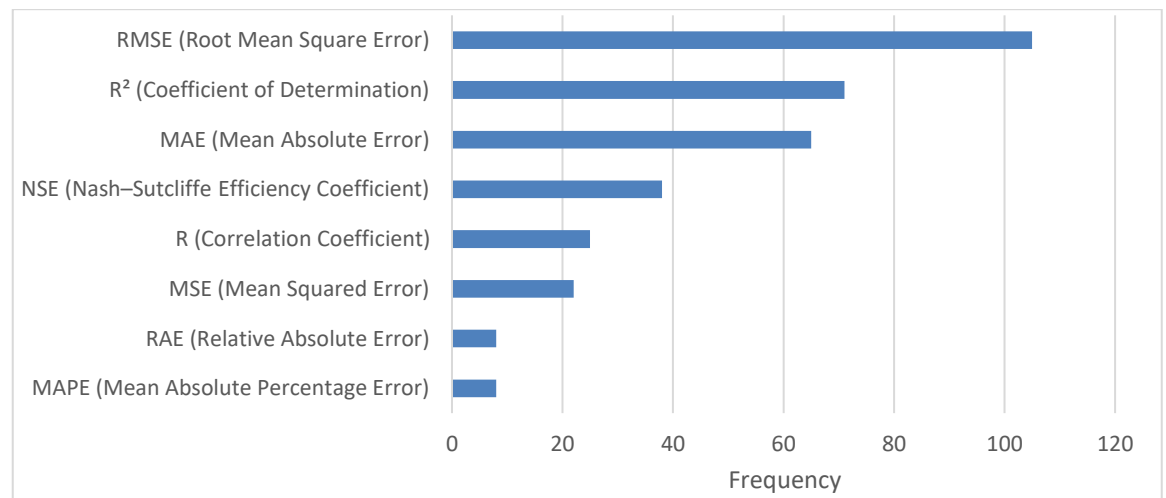


Figure 7. Main metrics used to evaluate groundwater prediction models

hydrogeological conditions, and prediction task. Even relatively simple architectures, such as ANN or radial basis function (RBF), can outperform more sophisticated models when combined with wavelet decomposition, as evidenced by [41], [44]. Hybrid models have also proven more robust when dealing with non-stationary or high-noise time series, a recurring phenomenon in monsoon regions of China, India, or Japan, where hydroclimatic variability is very high. In recent years, a shift in dominant architectures has emerged: while wavelet-ANN or RBF approaches initially prevailed, the current trend favors wavelet-LSTM [17], [45] along with a more recent move toward hybrid deep learning schemes such as CNN-LSTM models combined with ICEEMDAN or VMD [15], [46].

However, the review also reveals notable contradictions. One is the lack of consensus on which decomposition technique is most effective. Some studies conclude that wavelets deliver the best performance [37], [39], whereas others argue that EMD or VMD are better at handling noise and non-stationarity [17], [47]. Likewise, regarding the base architecture, while in certain cases wavelet enhanced ANN outperforms SVM [39], there are scenarios where non-neural models such as ANFIS or SVM become competitive or even superior after hybridization [40]. Limitations also appear with respect to the prediction horizon: while the benefits are consistent in the short term (weeks or months), results for long-term forecasts years ahead or under climate change scenarios are less robust or remain insufficiently explored.

3.2. Comparative studies of standalone artificial intelligence models

Another important line of research has focused on the direct comparison of different AI models to evaluate their relative performance in groundwater prediction. These studies aim to identify which architectures from classical ANN models (MLP, RBF) to more recent approaches such as LSTM, deep neural network (DNN), temporal convolutional network (TCN), or transformer models deliver the best results under varying temporal horizons and hydroclimatic conditions, as summarized in Table 2 [10], [18], [20], [34], [48]–[69] (see Appendix).

In the studies, a clear trend emerges toward the superior performance of more advanced models such as LSTM, DNN, or other recurrent variants compared to traditional neural networks like MLP or RBF, although important nuances appear depending on the prediction horizon and local conditions. Early investigations showed that hybrid approaches such as ANFIS and Bayesian neural networks could outperform simple ANN models [18], [48], while later comparisons highlighted the high accuracy of optimized MLPs, which in some cases achieved NSE values close to 0.99 [49], [54]. Starting in 2021, studies employing deeper architectures like LSTM and DNN [57], [58] began to demonstrate sustained advantages over classical models, particularly in capturing long-term and nonlinear temporal patterns. At the same time, works such as [10], [52] revealed that recurrent models like NARX remain highly competitive, especially when dealing with short or incomplete time series. In recent years, even more sophisticated architectures such as TCNs and transformers [34], [63] have consistently outperformed both LSTM and ANN across different prediction horizons, suggesting a shift toward next-generation deep learning models in this field.

Model performance depends on factors such as temporal scale, hydroclimatic variability, and data quality. For instance, in monsoonal regions of India and China, recurrent and deep models excel at handling high nonlinearity [57], [61], whereas in arid or semi-arid regions, studies with optimized ANN or ANFIS still report competitive results [55], [67]. Notable contradictions also emerge: in some cases, optimized ANN models achieve better metrics than ANFIS [62], while in others, ANFIS retains advantages under conditions of strong uncertainty [53], [68]. Another example lies in the comparison between statistical and neural models: recent work shows that Prophet or RF can reach similar performance levels [61], [66], indicating that higher model complexity does not always guarantee superior results.

3.3. Spatiotemporal prediction challenges

Another recurring theme in the literature focuses on the challenges of spatiotemporal prediction, where the main difficulty lies in overcoming the limitations of point-based data and capturing hydroclimatic variability across different temporal and spatial scales, as shown in Table 3 [70]–[78] (see Appendix).

An initial trend in spatiotemporal groundwater prediction studies using neural networks is the integration of satellite and meteorological data (e.g., GRACE, TRMM, and ERA-Interim), which helps compensate for the scarcity of local piezometric time series and improves performance even in simple architectures such as MLP [70], [71], [77]. A second trend is the move toward deeper and recurrent models, such as NARX and LSTM variants, which better handle long-term temporal dependencies and achieve higher accuracy levels [72], [74]. Overall, sequential models (LSTM, bidirectional long short-term memory (BLSTM), convolutional long short-term memory (ConvLSTM)) outperform classical ANNs, particularly when they incorporate satellite data [75], [78]. However, some contradictions remain: while the benefits are evident for short-term horizons, uncertainty increases for annual predictions [76], and there is no consensus on which spatial integration technique (e.g., clustering, interpolation) is most robust, as the results strongly depend on the hydrogeological context and the density of available data.

3.4. Quantitative synthesis of model performance

In quantitative terms, the reviewed studies report RMSE values ranging from 0.01 to 1.74 m, with LSTM models and their hybrid variants consistently achieving the lowest errors (in several cases, <0.1 m) and R^2 values exceeding 0.95, in some instances approaching 0.99. In contrast, classical ANNs (MLP, FFNN) displayed a broader and less stable performance range, with typical errors of 0.2–1.2 m and R^2 generally between 0.7 and 0.9. Hybrid models incorporating signal decomposition techniques (wavelet, EMD, and VMD) reduced errors by 20–40% compared to standalone neural network models, confirming the effectiveness of signal preprocessing.

3.5. Integration of data-driven models with the hydrogeological context

Although purely data-driven models have achieved high accuracy, it is equally important to understand how well they reflect the physical reality of the aquifer. Some studies have begun to incorporate parameters such as transmissivity, sediment thickness, recharge rates, or boundary conditions to enhance the model's physical plausibility. For example, Zaresefat *et al.* [79] considers soil transmissivity as a key variable for identifying optimal sites for artificial recharge. Similarly, Riseh *et al.* [80] includes sediment transmissivity and aquifer thickness among its inputs to predict subsidence linked to GWL decline. In India, Gomez *et al.* [81] highlights that model performance varies across sites depending on hydrogeological characteristics and monitoring well density. These examples show that, for predictions to be not only accurate but also truly useful for water resource management, it is essential to integrate the physical context of aquifers including, transmissivity, recharge, geological features, and boundary conditions.

3.6. Practical barriers to implementation

A key factor that conditions the applicability of neural networks in groundwater prediction is the minimum data requirement. In several studies, hybrid or recurrent models (such as LSTM) were trained using long time series: for example, Thakur *et al.* [82] employed 23 years of seasonal data in India for a PCA-DWT-LSTM model, while another study in Iran used 30 years of monthly observations to train an LSTM-NN for the Qazvin aquifer [83]. These cases show that, although some simpler models can work with shorter records, the ability to capture strong seasonal and interannual patterns improves significantly when longer series are available. When the temporal resolution is daily or hourly, as in coastal studies using LSTM, the data demand increases considerably [84]. Regarding data gaps and quality, techniques such as local interpolation and geostatistics combined with machine learning have been applied to fill missing values. For instance, [85] integrated machine learning with empirical Bayesian kriging (EBK) to predict monthly GWL anomalies.

3.7. Knowledge gaps

Common gaps have been identified that limit the consolidation of groundwater prediction using neural networks. First, most studies focus on traditional fit metrics (RMSE, NSE, R^2 , and MAE), but very few report uncertainty or robustness evaluations; only isolated cases, such as [15], include interval metrics (PICP and MPI). Second, there is a strong geographic concentration: studies are predominantly conducted in Asia (India, China, Iran, and Bangladesh), while analyses in arid or semi-arid environments of Latin America and Africa are limited, restricting the extrapolation of findings to other hydrogeological contexts. Another recurring gap is that most works focus on short prediction horizons (days to months), whereas multiyear projections or climate change scenarios are rarely explored, and when addressed, still show weak results (e.g., [76], with a decrease in accuracy over longer horizons). Finally, there remains a lack of systematic criteria for selecting architectures and preprocessing techniques: although significant improvements have been reported with wavelet, EMD, VMD, or meta-learning, the choice is usually based on trial and error (e.g., [47]), without clear guidelines to enable replication and transfer of models to different aquifers or data qualities.

4. CONCLUSION

This systematic review synthesized 150 studies published between 2009 and May 2025 and provides a global overview of neural network applications for groundwater prediction. The main contribution is the integrated analysis of predictive architectures, preprocessing strategies, evaluation metrics, and hydrogeological contexts. LSTM models and hybrid architectures frequently showed strong predictive performance, with several studies reporting RMSE below 0.1 m and R^2 above 0.95. Wavelet, EMD, and VMD, together with satellite observations, were also identified as promising strategies for improving prediction under data-limited conditions.

The review further shows that current evidence is constrained by limited integration of hydrogeological principles, dependence on long and high-quality time series, computational requirements, geographic concentration of studies, and insufficient evaluation of long-term and climate change scenarios. These limitations restrict the physical interpretability and transferability of existing models.

These findings have theoretical implications for groundwater prediction research by highlighting the importance of integrating neural network architectures, preprocessing strategies, and hydrogeological knowledge rather than relying solely on statistical performance. From a practical perspective, the findings can support the selection of prediction approaches according to data availability, hydrogeological conditions, and computational requirements.

Future research should prioritize hybrid and physics-informed approaches that integrate neural networks with hydrogeological knowledge, standardized evaluation protocols across multiple aquifers, uncertainty quantification, and explainability. Transfer learning, cross-regional validation, long-term forecasting, and computationally efficient architectures should also receive greater attention. Future empirical research should particularly focus on cross-regional validation of interpretable and physics-informed models using standardized evaluation protocols, especially for long-term groundwater prediction and climate change scenarios. These advances can improve the reliability and transferability of groundwater prediction models and strengthen their contribution to sustainable water resource management.

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AUTHOR CONTRIBUTIONS STATEMENT

This journal uses the Contributor Roles Taxonomy (CRediT) to recognize individual author contributions, reduce authorship disputes, and facilitate collaboration.

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C : **C**onceptualization

M : **M**ethodology

So : **S**oftware

Va : **V**alidation

Fo : **F**ormal analysis

I : **I**nterpretation

R : **R**esources

D : **D**ata Curation

O : **O**riginal Draft

E : **E**diting

Vi : **V**isualization

Su : **S**upervision

P : **P**roject administration

Fu : **F**unding acquisition

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY

Data availability is not applicable to this paper as no new data were created or analyzed in this study.

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APPENDIX

Table 1. Summarizes the main studies in this category, highlighting the types of networks used, the decomposition techniques applied, the study region, and the validation metrics

Author	Neural network type	Complementary technique	Region/country	Prediction frequency	Metric/evaluation method	Accuracy/error
[37]	Wavelet-ANN (WA-ANN)	Wavelet transform (WT)	Fujian, China	Monthly	R ² , RMSE, Nash-Sutcliffe efficiency (NSE)	RMSE: 0.12–0.14 m; NSE: 0.48–0.72; R ² : 0.99
[38]	Radial basis function neural network (RBFNN)	EMD	Jiansanjiang, Heilongjiang, China	Monthly	C, p, E1, E2, E3 (posterior error, small-error frequency, mean relative error, fit and prediction accuracy)	C=0.0871; p=1; E1=4.43%; E2=1; E3=100%
[39]	ANN (3-layer feedforward (5:25:1; activation function not specified; weights optimized with genetic algorithm (GA)) and WANN	SVM, wavelet support vector machine (WSVM), discrete wavelet transforms (DWT)	Mengcheng, China	Monthly	RAE, r (Pearson correlation), RMSE, NSE	WSVM: RAE=0.20, r=0.97, RMSE=0.18, NSE=0.94; WANN: RMSE=0.20, NSE=0.93; ANN lower
[40]	ANN (FFNN) (5:12:1; sigmoid hidden layer, linear output; trained with Levenberg-Marquardt), ANFIS, and SVM	Ensemble EMD (EEMD)	Florida, USA	Monthly	R, normalized mean squared error (NMSE), RMSE, NS, Akaike information criterion (AIC)	ANN: RMSE (validation)=0.329 (M1255); ANFIS: RMSE=0.360 (M1255); SVM: RMSE=0.315 (M1255)
[41]	FFANN and WANN (hybrid wavelet-ANN)	Gamma test, M-test, Wavelet coherence analysis	Konan Basin, Japan	Weekly (1, 3, 6, and 12 weeks)	RMSE, R, NSE, BIAS	Best results with WANN: e.g., E-2 (1 week): RMSE=0.33, R=0.95, NSE=0.90, BIAS=-0.056
[42]	ANN (3–7–1: 3 input nodes, 1 hidden layer with 7 neurons, 1 output node; sigmoid hidden layer and linear output); also LSSVM and NARX (nonlinear autoregressive network with exogenous inputs)	WT (db4)	Tasuj, Iran	Monthly	RMSE, R ²	Wavelet-NARX: R ² =0.99, RMSE≈0.01–0.14*
[43]	LSTM	EMD	Xinxiang, China	Monthly (2020–2021)	RMSE, relative error (min., max., and mean)	Mean relative error: 1.18%; max: 5.00%; min: 0.07%; accuracy: 100%
[44]	ANN MLP, tansig activation in hidden layer and linear output, learning rate 0.005, 5000 epochs, layers and neurons optimized via cross-validation	WT+PSR (WT-PSR)	Cangzhou, China	Every 10 days	RMSE, R (correlation), NSE	RMSE=0.295 m; R=0.999; NSE=0.995 (best case, well 2, dmey)
[45]	LSTM	WT, SHAP, remote sensing data (IMERG and MODIS)	Xi'an and Yinchuan, China	Monthly (1, 2, and 3 months)	NSE, RMSE, RSR, and Pearson (PR)	NSE=0.843 (1 month); RMSE≈0.203–0.250 m (best WT-LSTM values)
[46]	iTransformer	VMD	Kubuqi Desert, Hangjinqi, China	Every 4 hours (Jul 2021–Sept 2022)	MAE, RMSE, MAPE, and R ²	MAE: 0.0251; RMSE: 0.0262; MAPE: 1.2811%; R ² : 0.9287

Table 1. Summarizes the main studies in this category, highlighting the types of networks used, the decomposition techniques applied, the study region, and the validation metrics (*continued*)

Author	Neural network type	Complementary technique	Region/country	Prediction frequency	Metric/evaluation method	Accuracy/error
[47]	ANN (ReLU, 25 neurons in hidden layer); POA-ANN (ReLU, 10 neurons, optimized with Python optimizer algorithm); LSTM (50 neurons, tanh activation, +dense output layer)	126 preprocessing techniques (statistical, wavelet, VMD, and EMD)	Hamedan-Bahar Plain, Iran	Monthly	AIC, MAE, and RMSE	Best model: POA-ANN (ReLU, 10 neurons, MAE <1%); ANN (ReLU, 25 neurons) also notable
[15]	CNN-LSTM+LSTM	Improved complete EEMD with adaptive noise (ICEEMDAN), VMD, slime mould algorithm (SMA), adaptive weighting module (AWM) (adaptive fusion), SHAP (interpretability) EMD and WT	China (Jinan Baotu and Heihu Springs; Zhongtianshe Basin of Lake Taihu)	Daily/hourly	MAE, RMSE, MAPE, NSE, R ² , prediction interval coverage probability (PICP), and mean prediction interval (MPI)	RMSE: 0.0048 m (Baotu); 0.0100 m (Heihu); 0.0040 m (Taihu)/NSE: ~0.9998; MAPE <0.0001; PICP~1
[17]	LSTM		USA (California)	Daily prediction (multi-step)	RMSE, MAE, and R ²	EMD-LSTM: RMSE test W0804=0.1107; W0904=0.4489 (error reduction 19% and 78%)

Note: RSR, ratio of the RMSE to the standard deviation of observations.

Table 2. Summary of key studies in this category, highlighting the types of neural networks used, any applied decomposition techniques, the study region, and the validation metrics

Author	Neural network type	Complementary technique	Region/country	Prediction frequency	Metric/evaluation method	Accuracy/error
[18]	ANN.SCG, BNN.SCG, ANFIS	Scaled conjugate gradient (SCG), evidence approximation, and ANFIS	Dindigul, Southern India	Monthly (1 month ahead)	RMSE, relative error (RE), index of agreement (IA), and Pearson's R	RMSE (e.g., 83514): ANN: 1.27, Bayesian neural network (BNN): 0.66, ANFIS: 0.37; R (83514): ANN: 0.86, BNN: 0.94, ANFIS: 0.97
[48]	ANFIS, genetic programming (GP)	Iterative alternating direction implicit method (IADIM) (numerical simulation method)	Iran (Ghaen and Karaj)	Monthly	RMSE, MAE, Nash-Sutcliffe (E), R ²	Ghaen (testing): GP improves RMSE by 22.64%; Karaj (testing): GP improves RMSE by 22.75%; GP shows lower error and higher accuracy than ANFIS in both
[49]	MLP, RNN, RBF	GA	Konan, Japan	Monthly	RMSE, R ² , NSE, Bias, CV, and IOA	NSE: 0.828–0.998 (MLP-LM); Minimum RMSE: 0.02 m (H-2)
[50]	RBF (RBFNN)	SVM	Da'an, Jilin, China	Monthly	R, RMSE, MAE, NS, and d-factor	R ² RBF: 0.839 (training), 0.905 (validation); RMSE RBF: 0.315/0.413
[51]	ANN (MLP with LM)	GA–GT (GT+GA)	Hashtgerd, Iran	Monthly	R, RMSE, MAE, NSE, and developed discrepancy ratio (DDR)	Best case (SVR, 1 month, Well 2): R=0.97, RMSE=0.32, MAE=0.24, NSE=0.92

Table 2. Summary of key studies in this category, highlighting the types of neural networks used, any applied decomposition techniques, the study region, and the validation metrics (*continued*)

Author	Neural network type	Complementary technique	Region/country	Prediction frequency	Metric/evaluation method	Accuracy/error
[52]	NARX (RNN)	SVM	Mississippi, USA	Daily	MSE	NARX-ANN: MSE=0.01277; SVR: MSE=0.00123 (better)
[53]	BPNN (feedforward; 2 hidden layers; tanh activation in hidden, linear in output; trained with Levenberg–Marquardt backpropagation), ANFIS	GT and Pearson	Taiwan (Jhuoshui River Basin)	Daily	RMSE and Pearson correlation coefficient	Example G4(2): BPNN: RMSE=0.069, r=0.865; ANFIS: RMSE=0.062, r=0.89
[54]	NNARx (MLP)	Levenberg–Marquardt, Bayesian regularization, and early stopping	Italy (Tuscany, Monsummano Terme)	Daily	RMSE, MSE, sum of squared errors (SSE), NSE, Kling–Gupta efficiency (KGE), and AIC	Best model (Bayesian reg.): RMSE≈0.063 (5 days), NSE≈0.997 (1 day), very low AIC
[55]	GA-ANN with 8–9–1 architecture (8 input nodes, 1 hidden layer with 9 neurons, 1 output node; logistic sigmoid in hidden layer, and linear output)	GA–GA-ANN	India (Uttar Pradesh: between the Ganga and Hindon rivers)	Seasonal (pre- and post-monsoon)	R ² , coefficient of efficiency (CE), r, MAD, RMSE, coefficient of variation of residual errors (CVRE), absolute percentage error (APE), and performance index (PI)	GA-ANN-8 optimal model: R ² =0.94, RMSE=0.17 (testing, pre-monsoon)
[56]	FFNN (Feedforward), RBF (Radial basis)	Normalization, cross-validation	Najafabad, Iran	Monthly	RMSE and R ²	RMSE: 0.85–1.2 m; R ² : 0.92–0.97
[57]	ANFIS, DNN	SVM	India (18 agroecological zones)	Seasonal (4 times/year: Jan, Mar/Apr/May, Aug, Nov)	RMSE, percent bias (PBIAS), and NSE	DNN was most accurate (max NSE≈0.981), followed by ANFIS; SVM least accurate overall
[58]	MLP, LSTM, RNN, CNN	Hyperparameter optimization with RBF and GP (surrogate models)	Butte County, California, USA	Daily	MSE	MSE ≤0.002 (≤2 m error). MLP showed the best overall performance
[59]	MLP and RBF	BFGS and SCG (backpropagation algorithms)	Northern Algeria	Annual	RMSE and R ² (correlation coefficient)	MLP (BFGS 137) achieved RMSE=0.015, R ² >0.97 (excellent fit in training and testing)
[20]	ANN feed-forward backpropagation (FFBP) with variable architecture (1–X–1: 1 to 4 input nodes, 1 hidden layer, 1 output node; sigmoid hidden, linear output)	Xgboost, SVR, cross-correlation	Selangor, Malaysia	Daily	MAE, RMSE, R ² , and Willmott index (WI)	Xgboost: MAE=0.086, RMSE=0.137, R ² >0.92
[10]	NARX, LSTM, CNN	Bayesian optimization; TensorFlow/keras (LSTM); Python, MATLAB (CNN and NARX)	Upper Rhine Graben, Germany, and France	Weekly	NSE, R ² , RMSE, relative RMSE (rRMSE), Bias, relative bias (rBias), and PI	NARX best overall (NSE up to 0.9), robust with few data; CNN nearly as accurate but faster

Table 2. Summary of key studies in this category, highlighting the types of neural networks used, any applied decomposition techniques, the study region, and the validation metrics (*continued*)

Author	Neural network type	Complementary technique	Region/country	Prediction frequency	Metric/evaluation method	Accuracy/error
[60]	MLP and LSTM	Comparison with seasonal autoregressive integrated moving average (SARIMA) model	India (Punjab)	Biannual (pre- and post-monsoon)	RMSE, Pearson's correlation, and NSE	Minimum RMSE LSTM: 0.04; Max correlation: 0.99; Max NSE: 0.979 (MLP, pre-monsoon)
[61]	Generalized regression neural network (GRNN), LSTM, CNN, gated recurrent unit (GRU)	Shapley additive explanation (SHAP)	China (Tarim, Xinjiang)	Monthly (2000–2020; prediction: 2021–2023)	R ² , RMSE, PBIAS, and mean absolute relative error (MARE)	Best model: RF with average R ² =0.87; comparison: LSTM≈GRU; CNN lower than RF
[62]	ANN (3 inputs, 2 hidden layers with 6 neurons each, ReLU activation, 1 ReLU output neuron, Adam optimizer, 1250 epochs), ANFIS	Grid-search (ANN); Gaussian function (ANFIS)	Vidisha, India	Seasonal	MAE, MSE, RMSE, R ² -score, and NSE	ANN: MAE 0.85–1.04, R ² 0.77–0.83, NSE –1.25 to 0.98; ANFIS: MAE 0.90–1.18, R ² 0.76–0.81, NSE –1.27 to 0.99
[63]	TCN	Compared with LSTM and ANN; normalization; autocorrelation; dilated convolution	South Korea	1, 3, and 6 months	NSE, RMSE, and Pearson correlation	TCN outperformed LSTM (24%, 21%, 25%) and ANN (24%, 37%, 47%) in NSE at 1-, 3-, and 6-month horizons
[64]	CNN and LSTM	GAN for data imputation; univariate, Seq2val, Seq2seq	Taiwan (Choushui River Basin)	Daily	RMSE, MAE, R ² , SI, and BIAS	CNN (univariate): RMSE=0.007, R ² =0.998; LSTM (univariate): RMSE=0.008, R ² =0.997
[65]	RBFNN	Compared with stacked model (MLP+RF); Bayesian optimization	Grand Est, France	Daily (7, 15, and 30 days)	R ² , KGE, RMSE, MAE, MAPE, and relative error	RBF-NN outperformed MLP-RF in 2/3 wells (PZ and S1); R ² >0.92; MAPE between 1.66% and 10.01% (depending on well and horizon)
[66]	Prophet (additive time-series model)	GB, RF, SVR, DT, ANN	Rabat–Salé–Kénitra, Morocco	Daily (2010–2022)	MSE, RMSE, MAE, and R ²	MSE: 535.565, RMSE: 23.142, MAE: 16.949, R ² : 0.979
[34]	Transformer neural network	LSTM, CNN, MLP (comparative models)	Taiwan	10-day and 20-day	R ² and MAE	T+1: R ² =0.83, MAE=0.41 m; T+2: R ² =0.59, MAE=0.70 m
[67]	CNN, DNN	Interpolation with M5P, Kriging, and SVM	Azarshahr, Iran	Monthly	RMSE, R (correlation), and NSE	CNN: RMSE=0.888, R=0.996, NSE=0.984 (with 34 wells); DNN: RMSE=1.408, R=0.996, NSE=0.979
[68]	ANFIS	Ensemble BMA, COV, GRA, SE; Dempster-Shafer theory	Bangladesh	1, 2, and 3 weeks ahead	R, KGE, IOA, RMSE, MAE, MAD, and RSR	R=0.947; KGE=0.925; IOA=0.972; RMSE=0.123 m; MAE=0.062 m (for 1 week)
[69]	SVM	PCA	Yazd-Ardakan, Iran	Monthly (2012–2019)	RMSE, R ² , and AUC	RMSE: 0.12, R ² : 0.91, AUC: 0.94

Table 3. Summarizes the main studies in this category, highlighting the types of neural networks used, the decomposition techniques applied, the study region, and the validation metrics

Author	Type of neural network	Complementary technique	Region/country	Prediction frequency	Metric/evaluation method	Accuracy/error
[70]	MLP neural network (MLP, one hidden layer)	Gravity Recovery and Climate Experiment (GRACE) satellite data, PRISM (precipitation (Prcp) and temperature (Temp)), and ensemble technique	United States: Texas, Nebraska, Illinois (Test I); Nebraska (Test II)	Monthly (1 to 3 months ahead)	NSE, RMSE, MAE, correlation coefficient (R), and $\hat{R}$ (scaled RMSE)	NSE between 0.54 and 0.86; relative RMSE between 0.36 and 0.67 (Test I); similar performance in Test II with accuracy rated as “good” to “very good”
[71]	ANN (feedforward; input–hidden–output; layers/neurons and functions adjusted; trained with backpropagation and Rprop)	GRACE-DTWS+meteorological variables (temp, Prcp, wind, humidity, and lagged Prcp)	India	Quarterly (seasonal)	R^2 , q (correlation), and RMSE	R^2 between 0.3648 and 0.7428; q up to 0.9837 (train); minimum RMSE=0.0565 (train)
[72]	NARX (recurrent ANN)	Bayesian regularization (BR), MATLAB, autocorrelation function (ACF)/CCF	Barind Tract, Bangladesh	Weekly (52 weeks ahead)	MSE and R^2	R^2 between 0.79 and 0.99; minimum MSE≈0.008
[73]	NARX	Cross-correlation (XCF) and Bayesian regularization	Umbria, Italy	Weekly, biweekly, monthly	R^2 , RMSE, MAE, and RAE	$R^2=0.8864-0.9842$; RAE=0.0933–0.2557 (weekly resolution)
[74]	LSTM	Adam optimization	Texas, USA	Daily and monthly	R^2 , MSE, MAE	$R^2\geq 99.89\%$ (1 day), $\geq 99.00\%$ (4 days), $\geq 90\%$ (26 days)
[75]	ConvLSTM	K-means clustering (distance and hydrogeology), attention mechanism, and LASSO regression	Jinan, China	Every 5 days (6 times per month)	RMSE and NSE	Best model (M-MV-ConvLSTM): average RMSE=0.457, average NSE=0.216; in 50% of wells NSE>0.9 and in 80% NSE>0.85
[76]	Bidirectional long short-term memory (BLSTM), bidirectional highway long short-term memory (BHLSTM)+LSTM	Highway LSTM; inverse distance weighting (IDW) Interpolation (ArcGIS); normalization; dropout; Adam optimizer	Varuna River Basin, India	Annual (20 years of data)	RMSE and C-Index	Best model: 4 BHLSTM+LSTM; RMSE≈1.74 (test), C-index≈0.674; good long-term prediction
[77]	LSTM, ANN (feedforward backpropagation with 1–4 input neurons, 1 hidden layer with sigmoid activation, and 1 linear output)	GRACE, tropical rainfall measuring mission (TRMM), ERA-interim, global land data assimilation system (GLDAS)	Victoria, Australia	Monthly	PR (Pearson) and RMSE	LSTM: PR up to 0.863, RMSE 0.271 m; ANN: PR 0.519, RMSE 0.973 m
[78]	CNN-LSTM with meta-learning	Meta-learning (MAML), multivariate meteorological factors	Heihe, China (middle and lower basin)	Monthly	MAE and RMSE	Minimum MAE: 0.095 m (1 month), minimum RMSE: 0.135 m (1 month); 21.5%–44.6% improvement over classical methods

Note: DTWS: terrestrial water storage derived from GRACE observations; ACF: autocorrelation function; and CCF/XCF: cross-correlation function.

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